

A Multi-Functional VOC Sensor Based on Cascaded Quartz Crystal Resonators

Mengyao Fu, Zhiqiang Ma, Chenyang Gao, Yongping Ye, Wei Li, Dibo Hou, Yunqi Cao

Abstract—A multi-functional VOC (volatile organic compound) sensor composed of four parallel-connected quartz crystal resonators (QCRs) configured as a cascaded resonator was developed for VOCs identification and quantification. The structure of the cascaded resonator allows the high-resolution output response signals from the four QCRs to be obtained through a single measurement. Multi-parameter responses for each QCR can be calculated to form a hybrid sensor array, combining the advantages of multisensor array (MSA) and virtual sensor array (VSA), significantly enhancing the sensor's ability to identify both similar and diverse types of VOCs. Results show that the highest sensitivity is $16.14 \text{ Hz ppm}^{-1}$, with the lowest limit of detection is 0.34 ppm , exhibiting excellent sensing performance. Classification accuracy to eight VOCs is 98.68% of the hybrid sensor array. This development is of significant importance for the realization of artificial olfaction.

Index Terms—VOC detection, multisensor array (MSA), virtual sensor array (VSA), quartz crystal resonator (QCR), cascaded resonator.

I. INTRODUCTION

Volatile organic compounds (VOCs) pose significant threats to human health and environment [1]–[3], necessitating accurate detection methods [4]. With a quartz crystal resonator (QCR) as the core element, quartz Crystal Microbalance (QCM) is a representative gravimetric sensors with high precision at the nanogram level, offering superior performance in VOC sensing, where the adsorption layer significantly determine the sensing performance of the sensor [5]–[7]. Metal-Organic Frameworks (MOFs) possess a unique three-dimensional structure, high porosity, an ultra-large surface area, and abundant functional groups, which facilitate the efficient adsorption of VOC molecules, enabling QCM to detect VOCs with high sensitivity [8]–[11]. However, the similar physicochemical properties of VOCs make their precise differentiation challenging, for instance, distinguishing between acetone, ethanol, and methanol [12].

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This challenge spurred the development of electronic noses, or multisensor arrays (MSAs), featuring individual sensors coated with different sensitive materials, each responding to VOCs differently [13], [14]. However, the increased sensor units complicate the system. Furthermore, traditional MSAs require sequential measurements for each sensing unit, increasing the complexity and workload of data acquisition, and may require complicated circuitry [15]–[17]. To address these issues, virtual sensor arrays (VSAs) were introduced, employing a single physical sensor along with signal collection and data processing techniques for VOC identification [18], [19]. One approach to obtain sufficient response signals involves measuring QCM frequency signals at multiple harmonics, requiring high-precision equipment over a wide measurement range or multiple switches within the measurement frequency range [20]. The other approach involves parameter fitting of the Butterworth - Van Dyke (BVD) equivalent circuit model of the QCM, which requires significant time for data extraction [21]. Furthermore, reliance on only one sensitive material limits the chemical diversity of the sensing system, thus constraining performance when faced with more complex VOC classification tasks [21]–[23]. Therefore, it is urgent to develop a sensing device capable of providing rapid and convenient output responses, rich in chemical diversity, to accurately distinguish VOCs with similar physicochemical properties.

In this letter, we report a multi-functional VOC sensor based on the concept of cascaded resonator, achieved by paralleling QCRs of different frequencies. This design enables the acquisition of four resonant peaks with intervals that satisfy the 3 dB bandwidth and are relatively close, through a single frequency sweep ranging from 9.975 MHz to 10.0052 MHz. The detectable resolution is 3 Hz, allows the device to detect VOCs sensitively. Results show that the proposed device can detect VOC concentrations at parts-per-million (ppm) levels, and it can accurately identify various VOCs, achieving an average classification accuracy of 98.68% for the eight tested VOCs.

II. EXPERIMENTAL SECTION

The four QCRs coated with different MOF nanofilms MIL-101(Cr), UIO-66, ZIF-8, and PCN-222(H) in parallel (Fig. 1a) that constitute the MSA were fabricated following reference [24]. It can be observed from Fig. 1c-f that MIL-101(Cr), UIO-66, and ZIF-8 exhibit good crystalline structures [25]–[27], whereas PCN-222(H) displays a tubular structure [28],

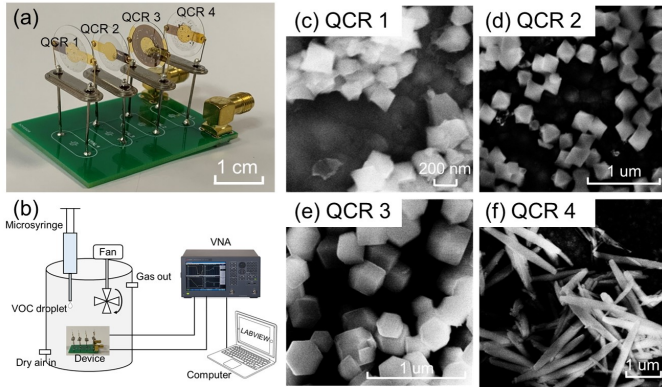


Fig. 1. (a) Physical diagram of VOC sensor. (b) Schematic diagram of experimental setup. Scanning electron microscope images of (c) MIL-101(Cr), (d) UIO-66, (e) ZIF-8, and (f) PCN-222(H) nanofilms.

facilitating the physical adsorption of VOCs. The resonant frequencies of the four coated-QCRs are 9.9809 MHz, 9.9906 MHz, 9.9957 MHz, and 10.0039 MHz. The experimental setup consists of the proposed sensor, a testing chamber, a vector network analyzer (VNA) with frequency scan range of 9.975-10.0052 MHz, and a computer (Fig. 1b). These resonant frequencies and the frequency scan range were chosen to ensure that the frequency difference between any two QCRs satisfies the 3 dB requirement, making each peak identifiable, as well as to enable the VNA to provide sufficient resolution [29], [30]. All experiments were conducted at room temperature (25°C) and under standard atmospheric pressure (1 atm = 101.325 kPa).

Before each VOC test, the chamber was purged with dry air (79% N₂ and 21% O₂). Subsequently, liquid droplets of 10-1000 ppm acetone, isopropanol, acetonitrile, toluene, methanol, ethanol, tetrahydrofuran, and chloroform (AC, IPA, ACN, TOL, MeOH, EtOH, THF, and CL), were introduced into the chamber. The Smith chart of each QCR near its fundamental resonant frequency (f_0) when exposed to VOC was measured by the VNA and recorded by LabVIEW (Fig. 1b). Smith charts (Fig. 2a) show scattering parameter in complex form, where the x-axis represents the real part (r) and the y-axis represents the imaginary part (x). The scattering parameter, resistance, reactance, impedance, and impedance angle (S_{11} , R , X , Z , and A) spectrum (Fig. 2b) can be derived through (1)-(4):

$$s_{11} = r + jx. \quad (1)$$

$$S_{11} = 20 \log_{10}(|s_{11}|). \quad (2)$$

$$|Z| = |50 \frac{1 + s_{11}}{1 - s_{11}}| = |R + jX|. \quad (3)$$

$$A = \arg(Z) = \tan^{-1} \frac{X}{R}. \quad (4)$$

Using the shifts of f_0 , S_{11min} , R_{max} , X_{max} , Z_{max} and A_{max} , construct the 6-dimensional (6-D) output for the VSA (Fig. 2b). It should be noted that, although these parameters may seem redundant, introducing more of them into the classification algorithm can increase the model's tolerance to deviations in individual or a few parameter values (since

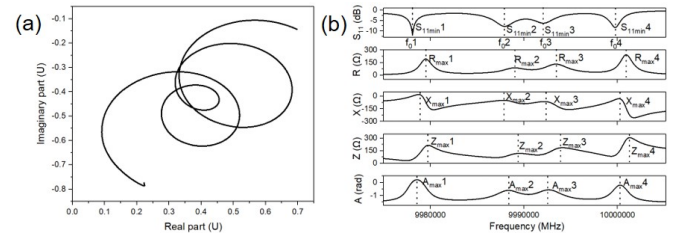


Fig. 2. (a) Smith chart, (b) S_{11} , R , X , Z , A spectrum of the cascaded resonator.

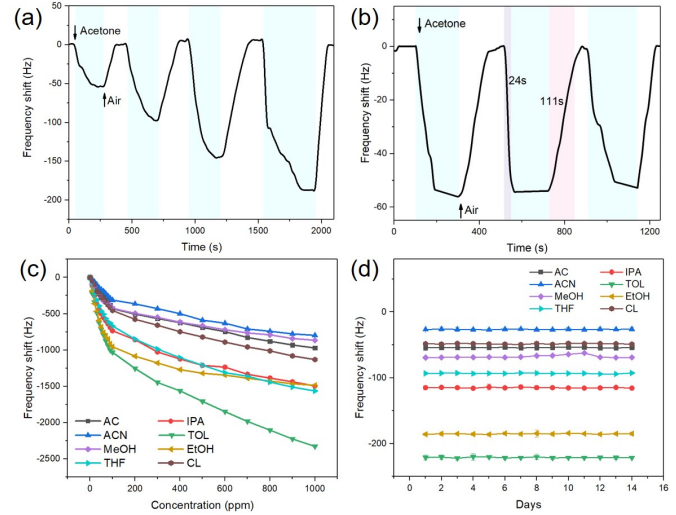


Fig. 3. (a) Dynamic response curve to 10-40 ppm acetone, (b) repeatability response curve to 10 ppm acetone, (c) sensing performance, and (d) long-term stability curve to eight VOCs of Sensor 1.

these parameters are derived from modeling and may deviate from actual conditions), thereby reducing the reliance on data accuracy. Additionally, incorporating these parameters into the algorithm can prevent the algorithm from learning their patterns on its own, which reduces the algorithm's complexity and improves its learning efficiency, ultimately enhancing overall classification performance. In this study, the classification accuracy improved by approximately 20%. Consequently, the 4 QCRs provided a 24-dimensional (24-D) output for the hybrid sensor array.

Support Vector Machine (SVM) was employed to identify eight VOCs and the leave-one-out cross-validation (LOOCV) was used to calculate recognition accuracy. The LOOCV leaves one sample for the test set at a time, while leaving others for the training set, resulting in a series of models, and the final assessment is the average of all test results.

III. RESULTS AND DISCUSSION

To investigate the sensing performance of the MSA, the f_0 (Fig. 3b) shifts of each QCR were recorded when switching from dry air to environments containing different concentrations of specific VOCs, serving as the sensing signal of the MSA. Responses to eight VOCs at various concentrations (10-100 ppm in 10 ppm increments, 100-1000 ppm in 100 ppm increments) for AC, IPA, ACN, TOL, MeOH, EtOH, THF, and CL were explored. Fig. 3a illustrates the dynamic response

TABLE I
THE SENSITIVITY (Hz ppm⁻¹) OF EACH QCR TO DIFFERENT VOCs

VOC	QCR 1	QCR 2	QCR 3	QCR 4
Acetone	4.15	0.62	0.42	0.38
Isopropanol	11.18	3.00	0.44	0.61
Acetonitrile	3.17	0.56	0.46	0.61
Toluene	16.14	4.00	0.31	0.15
Methanol	5.10	0.40	0.46	0.42
Ethanol	15.60	1.99	0.19	0.44
Tetrahydrofuran	7.60	2.18	1.11	1.10
Chloroform	4.52	0.83	0.41	0.29

curve of QCR 1 (i.e., MIL101-coated QCR) to acetone (10–40 ppm), with decreasing f_0 as VOC concentration increases. Fig. 3b presents the repeatability of Sensor 1 under 10 ppm acetone conditions in 3 cycles, with response and recovery times of 24 and 111 seconds, respectively. The frequency shifts served as a function of VOC concentrations (Fig. 3c, QCR 1 for example), with the slope within the linear range indicating sensitivity (Table 1), ranging from 0.19–16.14 Hz ppm⁻¹. The data measurements were conducted 11 times at each VOC concentration, and the average relative standard deviation was calculated to be 0.43%, indicating good repeatability of the sensing performance. Notably, each sensor exhibits different sensitivities due to varying MOF film absorption characteristics and different amounts (or thicknesses). For all VOCs, the MIL-101(Cr)-coated QCR shows the highest sensitivity, likely due to its highest specific surface area [31]. Different VOCs also present varying affinities within the same MOF. For instance, MIL-101(Cr) has the highest affinity for toluene [32], while ZIF-8 favors tetrahydrofuran [33]. The limit of detection (LOD) for different VOCs was calculated to be between 0.34–1.79 ppm, based on the general criterion of a signal-to-noise ratio greater than 3 [34]. Additionally, the sensing responses to 10 ppm VOCs over 14 days (Fig. 3d) showed a small coefficient of variation (3.01%) under identical concentrations, indicating good stability.

Furthermore, a VSA based on each QCR was constructed to build multidimensional fingerprints for each VOC. Combining shifts of f_0 , S_{11min} , R_{max} , X_{max} , Z_{max} , and A_{max} , each QCR can provide a 6-D VSA, resulting in a 24-D feature vector for the hybrid sensor array. Notably, due to the implementation of the cascaded resonator, the 24-D features require only one data acquisition, significantly enhancing data collection efficiency. Eight different VOCs were used to evaluate the classification abilities of MSA, VSA, and the hybrid sensor array. Linear discriminant analysis (LDA) was performed for multivariate statistical analysis (Fig. 4a–c), verifying their classification capabilities for VOCs. Confusion matrices (Fig. 4d–f) were used to visually reflect the classification performance, with correct classifications appearing along the blue diagonal line. The classification accuracies are 82.89% (MSA), 90.79% (VSA), and 98.68% (the hybrid sensor array). MSA achieves 78.95% for alcohol substances and 85.25% for non-alcohol substances. VSA of Sensor 1 achieves 89.47% for alcohol substances and 91.58% for non-alcohol substances. This indicates that MSA, due to its rich chemical diversity, performs better in classifying different

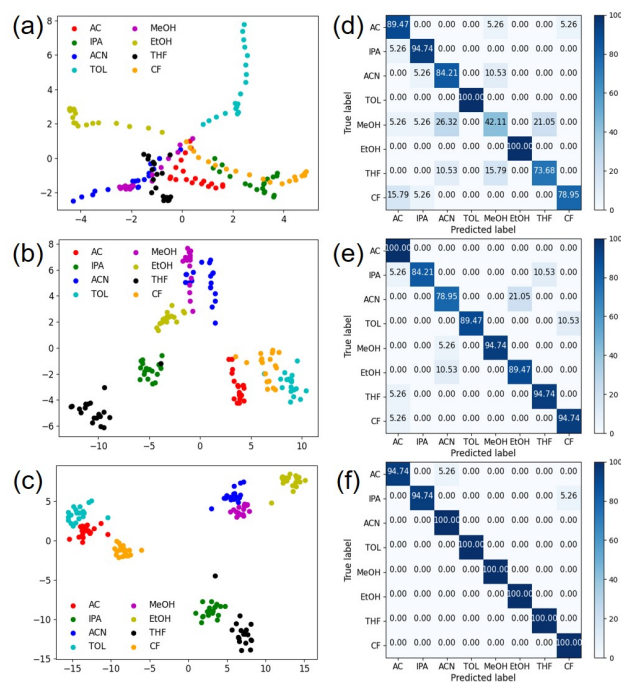


Fig. 4. The identification of VOCs by LDA plots in 2-dimension of (a) MSA, (b) VSA, (c) the hybrid sensor array. The confusion matrices for VOCs classification using SVM of (d) MSA, (e) VSA, (f) the hybrid sensor array.

types of VOCs, whereas VSA, by leveraging the viscoelastic properties of sensitive films, focuses on interactions between the film and VOC molecules, can perform better than MSA in classifying similar substances [35]. The hybrid sensor array, however, combines both advantages, excelling in recognizing similar and diverse substances.

IV. CONCLUSION

In this letter, a multi-functional VOC sensor was designed and developed by implementing the concept of cascaded resonator. Leveraging the cascaded resonator approach, the data acquisition process has been optimized to generate high-resolution output signals from a single data collection event for the quantitative detection of VOCs, with a sensitivity range of 0.19 to 16.14 Hz ppm⁻¹ and the LOD is between 0.34–1.79 ppm, indicating promising application prospects. Additionally, a hybrid sensor array was constructed based on these signals for the qualitative classification of eight VOCs. Experimental results demonstrate that the hybrid sensor array can enhance the functionalities of existing MSA and VSA, boasting accurate identification and quantification capabilities for eight categories of VOCs, with an accuracy rate of 98.68% while 82.89% of MSA and 90.79% of VSA. In future work, the classification and sensing performance of the device under gas mixture exposures, as well as methods for large-scale production of integrating MOF nanofilms onto QCR, will be explored.

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